

American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

PAW Connect Dhaka

Semester: Fall 25-26

Group: 7		***Section: F***
SL	**Student Name**	**Student ID**
1	Moynul Islam Mahi	23-50350-1
2	Mahfuzur Rahman Khan Fahad	23-50349-1
3	MD. Jahidul Islam	23-50365-1
4	Sanzida Aruba Shorony	23-50367-1
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Table of Contents

1. Project Proposal 03
   1.1 Background to the Problem 03
   1.2 Selection of Process Model 07
2. Software Requirements Specification (SRS) / PRD 10
   2.3.2 Non-Functional Requirements 18
3. Software Design 21
   3.1 System Deisgn 21
      3.1.1 Use case Diagram 22
   3.2 UI Deisgn using Figma 24
4. Git Workflow 30
5. Software Testing 31
6. Conclusion xx
7.

1. PROJECT PROPOSAL

1.1 Background to the Problem

In the last few years, the count of abandoned, lost, or homeless pets has grown rapidly. There are many kind-hearted people who are willing to adopt animals, but there is no proper system connecting adopters, pet owners, shelters, and clinics in one trusted digital space.

Adoptions are taking place through postings on social media, miscellaneous websites, and personal networking. Such an informal approach has created problems relating to fake listings, scams, unverified information relating to pets, and waiting times. Owners are struggling to find adopters, while shelters are filled with animals because of the low visibility of adoptions.

At the same time, people looking to adopt find it difficult to locate pets nearby, check their health status, or connect safely with the owner. There is also a lack of awareness about legal adoption procedures and post-adoption training. Moreover, there is no organized system to help with lost and found pets, community discussions, or locating nearby pet clinics and shelters.

Root cause of the problem

1. No centralized platform dedicated to pet adoption and welfare.
2. Lack of verified user identities and transparent communication.
3. No proper database of nearby shelters, clinics, and events.
4. Absence of awareness and updates on legal adoption guidelines.
5. Rising incidents of fraud and fake pet listings.

Why this problem is so important?

1. **Animal Welfare:** Many pets remain homeless or end up in unsafe conditions.
2. **Safety & Trust:** Adopters face scams and misinformation due to unverified platforms.
3. **Convenience:** Pet owners and adopters waste time searching across different sources.
4. **Community Impact:** Missing pets, illegal trades, and untrained adoptions harm both animals and communities.
5. **Digital Gap:** Shelters and clinics often lack online presence and visibility.

1.1.1 Solution of the problem:

Objective:

The goal is to develop a smart, user-friendly web and mobile platform for adopters, owners, shelters, and pet care providers that ensures safe, transparent, and location-based adoptions with community support and fraud prevention.

Solution:

We are going to develop a **Pet Adoption Platform** where:

1. Pet owners, shelters, and NGOs can **list pets** for adoption with verified details.
2. Adopters can **search and filter pets** by category, breed, age, or distance.
3. Both parties can **chat securely** through an in-app **messaging system**.
4. The system will provide **location-based services**, showing nearby shelters, pet homes, clinics, and pet-related events.
5. A **live community** will help users share tips, report lost pets, and get support.
6. **Fraud alert and rating systems** will ensure trust and safety.
7. **Legal guidelines and pet care training** will be updated daily for awareness.

Why this works?

1. Makes adoption faster and safer through verified listings.
2. Builds a connected ecosystem between pet lovers, shelters, and clinics.
3. Reduces fraud through rating-based alerts and verified users.
4. Encourages responsible pet ownership and adoption awareness.
5. Works for all age groups and skill levels through a simple, accessible interface.

Feasibility:

1. Available on both **web and mobile devices** (Android/iOS).

2. Uses **real-time location** and **smart search** for better matching.
3. Revenue can be generated through **ads, donations, or premium shelter listings.**
4. Simple interface for all users, including non-technical ones.
5. Easily scalable for use in different cities or countries.

Key Features:

1. Pet Listing (Category-wise)
2. Smart Search
3. Adoption Request System
4. User Info & Verification
5. Message System (Between Adopter and Owner)
6. Location Finder (Nearby Shelters, Clinics, Events)
7. Live Support Community
8. Lost & Found Pet Feature
9. Legal Agreement & Daily Training Updates
10. Fraud Alert System (Rating-Based)

Target Users:

1. **Pet Owners:** Wanting to rehome pets responsibly.
2. **Adopters:** Looking for verified pets to adopt.
3. **Animal Shelters & NGOs:** Managing adoption and welfare programs.
4. **Pet Clinics:** Offering healthcare and checkup services.
5. **Pet Enthusiasts:** Participating in community activities and discussions.

Benefits:

1. Simplifies and speeds up the pet adoption process.

2. Promotes safe and verified adoption.
3. Helps reunite lost pets with owners.
4. Builds a supportive pet-loving community.
5. Reduces fraudulent activities and misinformation.
6. Connects shelters and clinics with more people in need.

Research Support

Similar platforms:

1. **Petfinder (USA)** – Focuses on shelter adoptions but lacks local community features.
2. **AdoptAPet (Global)** – Great listings but limited real-time communication.
3. **Animal Rescue Network (Bangladesh)** – Uses Facebook groups, no verification or fraud prevention.

How We Improve Existing Solutions:

Platform	Problems	Our App Adds
Facebook Groups	Unverified, scattered posts	Verified listings, smart search, fraud alerts
Petfinder	No community or real-time chat	Adds messaging, lost & found, live support
AdoptAPet	Focused on shelters only	Includes owners, adopters, clinics, and events
Local NGOs	Manual process, limited reach	Centralized system, automated alerts

We focus on:

1. Verified, Safe, and Transparent Adoption
2. Location-Based Search and Matching
3. Community Awareness and Support
4. Legal and Training Information
5. Lost & Found and Fraud Prevention System

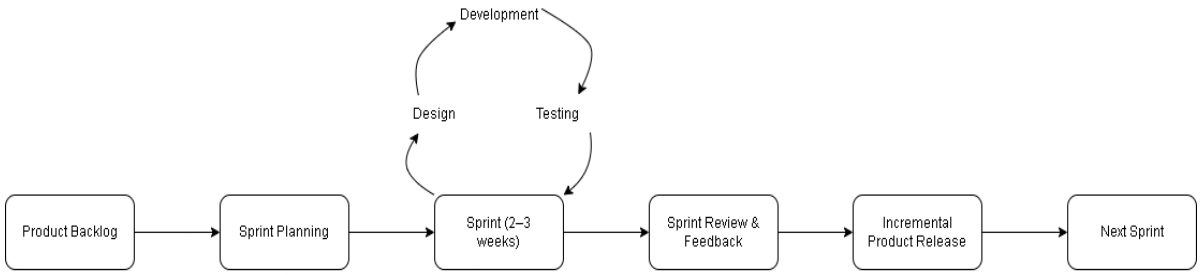
1.2 Selection of Process Model

For the development of the **PawConnect Dhaka** project, the **Agile (Scrum) Process Model** has been selected..

The Agile model is an **iterative and incremental software development approach** where the project is divided into small development cycles known as **sprints**. Each sprint typically includes the phases of:

- Requirement analysis
- Design
- Development
- Testing
- Review

Instead of developing the entire system at once, features are developed, tested, and delivered **incrementally**, allowing continuous improvement and feedback.



Agile (Scrum) Workflow Visualization

Each sprint results in a **working version of the software**, which can be evaluated and enhanced in the next iteration.

1.2.1 Why Agile Model Was Chosen for This Project

The Agile model is particularly suitable for **PawConnect Dhaka** due to the following reasons:

1. **Evolving Requirements**

   The project includes multiple features such as authentication, pet management, messaging, donations, and admin analytics. Agile allows requirements to be **refined or extended** during development without disrupting the entire system.

2. **Feature-Based Development**

   The system can be developed module by module (e.g., Authentication, Pet Management, Adoption, Donation), which aligns well with Agile's **incremental delivery** approach.

3. **Frequent Feedback**

   Regular sprint reviews allow stakeholders and team members to provide feedback early, helping improve usability and functionality before final deployment.

4. **Risk Reduction**

   High-risk features such as **payment gateway integration, fraud reporting, and verification systems** can be developed and tested early in separate sprints, reducing overall project risk.

5. **Improved Team Collaboration**

   Agile encourages continuous communication and collaboration among team members, making it ideal for a **team-based academic software engineering project**.

6. **Faster Delivery of Core Features**

   Essential features like login, pet listing, and adoption requests can be delivered early, while advanced features like analytics and policy management can be added in later phases.

1.2.2 Conclusion

Considering the overall scope of the project, its multi-module architecture, and the involvement of diverse user roles, the **Agile (Scrum) Process Model** is the most suitable approach for developing *PawConnect Dhaka*. The system includes several interconnected features such as authentication,

pet management, adoption workflows, community interaction, payment integration, and administrative controls, which require frequent refinement and validation. Agile supports **incremental development**, allowing features to be designed, implemented, tested, and reviewed in short iterations. This enables the development team to respond effectively to changing requirements, incorporate continuous feedback, and reduce technical and functional risks. Moreover, Agile facilitates efficient management of both **functional and non-functional requirements** by promoting regular evaluation, improved collaboration, and continuous improvement throughout the entire software development lifecycle.

2. SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS) / PRODUCT REQUIREMENTS DOCUMENT (PRD)

The PRD was created based on stakeholder needs and system analysis, and it aligns with the selected Agile development model.

Template Name	PawConnect Dhaka – Smart Pet Adoption & Care Platform
Goals & Objective	Centralize pet adoption and welfare in one trusted system. Prevent fake listings and fraud using user verification and reporting. Promote responsible pet ownership and awareness through community features. Integrate real-time location-based discovery for pets, clinics, and shelters.
Client Information	
Product Type	Web & Mobile (Android/iOS)
Domain	Web2
Access Layers	Admin, Pet Owner[User], Adopter[User], Shelter/NGO[User], Clinic/Vet[User], Community Member[User]
Design Required	Yes

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 1		**Authentication & Access Control**				
phase 1	1	Signup	<Form> <input=name> <input=email> <input=gender> <input=dob> <input=password> <input=re-password> <input=user-type> <button=signup> <link=signin>	As a new user, I want to sign up with my details so I can create an account and access the platform.	• User must provide valid name, email, password, and user type. • System checks for existing email before signup. • Confirmation message displayed after successful registration.	Signup
phase 1	2	Login	<Form> <input=email> <input=password> <option=remember me> <button=signin> <link=signup> <link=forgot password>	As a user, I want to log in securely so I can access my personalized dashboard.	• Login validates credentials correctly. • Invalid credentials show error message. • User is redirected to the dashboard upon success.	Login
phase 1	3	Logout	<Page> <button=logout> <message=Are you sure you want to logout?> <button=yes> <button=cancel>	As a user, I want to log out so that my session ends safely.	• Clicking logout clears all user session data. • User is redirected to the login page.	Logout
phase 1	4	Forgot Password	<Form> <input=email> <button=send code> <input=code> <input=new password> <input=confirm password> <button=submit>	As a user, I want to reset my password if I forget it.	• System sends a verification code to registered email. • User can reset password after verifying the code. • Password updates successfully and old one becomes invalid	ForgotPass

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 2		**User Verification**				
phase 1	5	Document Upload	<Form> <input=user-id> <upload=ID document> <dropdown=user-type> <button=submit verification>	As a user, I want to upload my ID to verify my account.	• System accepts image/PDF format uploads. • Upload progress and preview displayed before submission.	VerifyDoc
phase 2	6	Admin Approval	<Page> <list=pending verification requests> <button=approve> <button=reject> <label=status updated>	As an admin, I want to approve or reject verification requests.	• Admin can view all pending verifications. • On approval, the user receives a "Verified" badge.	VerifyAdmin
phase 2	7	Verified Badge Display	<Page> <display=user profile> <icon=verified badge next to username>	As a verified user, I want my profile to show a verification badge.	• Verified badge visible beside name in listings and profile. • Badge updates automatically upon verification.	Badge

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 3		**Pet Management**				
phase 1	8	Add Pet	<Form> <input=pet name> <dropdown=category> <dropdown=breed> <input=age> <input=health condition> <textarea=description> <upload=pet photo> <button=add listing>	As a pet owner, I want to list a pet for adoption with full details	• System validates all required fields (name, category, breed, etc.). • Pet appears in the public listing upon submission.	AddPet
phase 1	9	Edit Pet	<Form> <select=existing pet> <edit=fields (name, age, health, photo)> <button=save changes>	As a pet owner, I want to edit my pet's details if any information changes.	• Editable fields include name, health, and photo. • Updates reflect immediately in listings.	EditPet
phase 1	10	Delete Pet	<Page> <list=user pet listings> <button=delete> <confirm=Are you sure?>	As a pet owner, I want to delete my pet listing when it's adopted.	• Deletion prompts confirmation message. • Listing removed from public view instantly.	DelPet

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 4		**Dashboard & Navigation**				
phase 1	11	Dashboard Overview	<Page> <section=user stats> <widget=my pets> <widget=adoption requests> <widget=messages> <widget=events>	As a user, I want a dashboard showing my activities and quick links.	• Dashboard displays pet count, requests, and messages. • Links navigate to correct pages.	Dashboard
phase 1	12	Menu Navigation	<Menu> <link=home> <link=dashboard> <link=messages> <link=profile> <link=settings> <link=logout>	As a user, I want an easy-to-navigate menu.	• Menu links (Home, Profile, Messages, Settings) open correct screens. • Navigation works consistently across all pages.	Menu

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 5		**Smart Search & Filter**				
phase 1	13	Pet Search	<Form> <input=search keyword> <button=search> <list=search results>	As an adopter, I want to search for pets by name or breed.	• Search returns relevant pets instantly. • Displays message if no match is found.	Search
phase 1	14	Filter Options	<Form> <dropdown=category> <dropdown=breed> <dropdown=age> <dropdown=location> <button=apply filter> <button=clear>	As an adopter, I want to apply filters to narrow down search results.	• Filters by age, breed, distance, and category. • Combined filters refine results dynamically.	Filter

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 6		**Adoption Management**				
phase 1	15	Send Request	<Form> <display=pet details> <input=message to owner> <button=send request>	As an adopter, I want to send an adoption request to a pet owner.	• Request stored with timestamp and user info. • Owner notified instantly.	SendReq
phase 1	16	Request Status Tracking	<Page> <list=sent requests> <label=status (pending/approved/rejected)> <button=cancel request>	As an adopter, I want to track my request status.	• Status updates automatically (Pending → Approved/Rejected). • User notified on status change.	ReqStatus

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 7		**Messaging & Communication**				
phase 2	17	Chat Interface	<Page> <list=recent chats> <display=chat messages> <input=type message> <button=send>	As a user, I want to chat with others to discuss adoption details.	• Messages delivered instantly and stored in history. • Chat available only between verified users.	Chat
phase 2	18	Media Sharing	<Form> <upload=image/video> <button=send media>	As a user, I want to send pet photos in chat.	• Images upload and display in message thread. • File size limited for performance.	Media

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 8		**Location & Discovery**				
phase 2	19	Map View	<Page> <map=google map> <pin=shelters> <pin=clinics> <pin=events> <button=view details>	As a user, I want to see nearby shelters and clinics on a map.	• Pins display correct locations. • Clicking a pin opens details.	Map
phase 2	20	Directions	<Page> <display=selected location> <button=get directions> <map=route preview>	As a user, I want to get directions to a shelter or clinic.	• Clicking "Get Directions" opens map navigation.	Direction

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 9		**Lost & Found System**				
phase 2	21	Report Lost Pet	<Form> <dropdown=Lost> <input=pet name> <upload=photo> <input=last seen location> <textarea=description> <button=submit>	As a user, I want to post details of my lost pet.	• User uploads photo, location, and description. • Report appears on community map.	LostPet
phase 2	22	Report Found Pet	<Form> <dropdown=Found> <upload=photo> <input=found location> <textarea=description> <button=submit>	As a user, I want to report a found pet to help reunite it with its owner.	• Found report includes location and image. • Owner notified if pet matches description.	FoundPet

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 10		**Fraud Detection & Reporting**				
phase 2	23	Submit Report	<Form> <dropdown=report type> <input=listing/user id> <textarea=reason> <upload=screenshot optional> <button=submit>	As a user, I want to report fake or unsafe listings.	• Report form includes reason and optional screenshot. • Admin notified instantly.	FraudReport
phase 3	24	Admin Action	<Page> <list=reported items> <button=mark resolved> <button=suspend account>	As an admin, I want to review and act on fraud reports.	• Admin can suspend or delete listings. • Report marked as resolved after action.	FraudAct

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 11		**Rating & Review System**				
phase 2	25	Add Review	<Form> <input=rating (1–5)> <textarea=comment> <button=submit review>	As an adopter, I want to leave feedback on an adoption experience.	• Review includes rating and text comment. • Only verified users can review.	AddRev
phase 2	26	Display Reviews	<Page> <list=reviews> <display=average rating> <filter=by date or rating>	As a user, I want to see ratings and reviews on profiles.	• Average rating calculated dynamically. • Reviews sorted by newest first.	ViewRev

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 12		**Profile Management**				
phase 1	27	View Profile	<Page> <display=profile photo> <display=name> <display=email> <display=user type> <display=rating> <list=my listings>	As a user, I want to view my personal details and adoption history.	• Displays user info, listings, and ratings.	ViewPro
phase 1	28	Edit Profile	<Form> <input=name> <input=phone> <input=email> <textarea=bio> <button=save>	As a user, I want to edit my contact info or bio.	• Changes save instantly after validation.	EditPro
phase 1	29	Change Profile Photo	<Form> <upload=new photo> <preview=image> <button=update photo>	As a user, I want to update my profile picture.	• New photo preview shown before saving.	ProPhoto
phase 1	30	Change Password	<Form> <input=current password> <input=new password> <input=confirm password> <button=update password>	As a user, I want to change my password for security.	• System verifies old password before update.	ChgPass

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 13		**Community & Awareness**				
phase 2	31	Post Discussion	<Form> <input=post title> <textarea=content> <upload=image optional> <button=publish>	As a community member, I want to share posts about pet care.	• Posts can include text or images.	Post
phase 2	32	Comment & Reply	<Form> <input=comment> <button=reply> <list=replies>	As a user, I want to comment on community posts.	• Comments appear instantly and are editable by author.	Comment

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 14		**Events & Campaigns**				
phase 3	33	View Events	<Page> <list=upcoming events> <display=event details> <button=view map>	As a user, I want to view all upcoming pet adoption events.	• Event list shows date, time, and organizer.	EventView
phase 3	34	Register for Event	<Form> <display=event info> <button=register> <confirmation=registration successful>	As a user, I want to register for an event.	• Registration confirmation sent via email.	EventReg

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 15		**Admin Panel & Analytics**				
phase 3	35	Manage Users	<Page> <list=all users> <button=verify> <button=ban> <button=delete> <filter=by role>	As an admin, I want to view and manage all user accounts.	• Admin can ban, verify, or delete users.	MngUser
phase 3	36	View Analytics	<Page> <chart=adoption trends> <chart=active users> <chart=reports summary> <button=export CSV> <button=export PDF>	As an admin, I want to view platform analytics.	• Graphs show adoptions, active users, and reports. • Reports exportable as PDF/CSV.	Analytics

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 16		**Donation & Sponsorship**				
Phase 3	37	User Donations	<Form> <dropdown=select shelter> <input=amount> <dropdown=payment method> <button=donate> <message=donation successful>	As a user, I want to donate money to shelters so I can support animals in need.	• User can select shelter and donation amount. • System processes payment securely. • User receives confirmation receipt.	Donate
Phase 3	38	Pet Sponsorship	<Form> <display=pet details> <dropdown=sponsor type> <input=amount> <button=sponsor> <message=sponsorship confirmed>	As a user, I want to sponsor a specific pet so I can help with its care.	• User can select a pet profile to sponsor. • System shows monthly or one-time sponsor options. • Sponsorship status updates automatically.	Sponsor
Phase 3	39	Payment Gateway Integration	<Page> <dropdown=payment method> <input=card/bkash info> <button=pay now> <message=payment success/fail>	As a user, I want a secure payment method to complete donations or sponsorships.	• Supports card/Bkash/Nagad/etc. • Payment success/failure messages shown. • Transaction stored in database.	Payment

Phase	Item No	Specification [Role]	Screen Definition	User Story	Acceptance Criteria	Tags
Feature: 17		**Legal & Policy Management**				
Phase 3	40	Adoption Agreement	<Page> <display=agreement content> <checkbox=I agree> <button=sign agreement> <message=agreement signed successfully>	As an adopter, I want to read and sign an adoption agreement so the adoption process is legally valid.	• Agreement auto-generated with pet and adopter info. • User can sign digitally. • Copy stored and downloadable.	Agreement
Phase 3	41	Terms Update	<Form> <textarea=terms content> <button=publish update> <message=terms updated>	As an admin, I want to update platform terms so the system remains compliant with policies.	• Admin can edit policy or terms text. • New version saved and published. • Users notified about updates.	Terms
Phase 3	42	Policy Notifications	<Popup> <display=new policy update> <button=ok/acknowledge>	As a user, I want to receive policy notifications so I stay informed about important updates.	• Notification shown in dashboard when terms change. • User must acknowledge updated policies. • Notification disappears after acknowledgement.	PolicyNote

2.3.2 Non-Functional Requirements

NFR ID	Category	Requirement Description	Priority	Verification Method	Related Feature / Module	Phase
NFR-01	Performance	The system shall respond to user actions within 2 seconds under normal load.	High	Performance Testing	Dashboard, Search	Phase 1
NFR-02	Performance	The system shall support multiple concurrent users without degradation.	High	Load Testing	Authentication, Messaging	Phase 2
NFR-03	Performance	Search and filter results shall load efficiently with minimal delay.	High	Performance Testing	Search & Filter	Phase 1
NFR-04	Security	User passwords shall be encrypted before storing in the database.	High	Security Review	Authentication	Phase 1
NFR-05	Security	The system shall implement role-based access control (User/Admin).	High	Inspection	User Management	Phase 1
NFR-06	Security	Payment transactions shall be processed using secure payment gateways.	High	Security Testing	Payment, Donation	Phase 3
NFR-07	Security	Uploaded documents and images shall be protected from unauthorized access.	High	Security Audit	Verification, Media	Phase 2

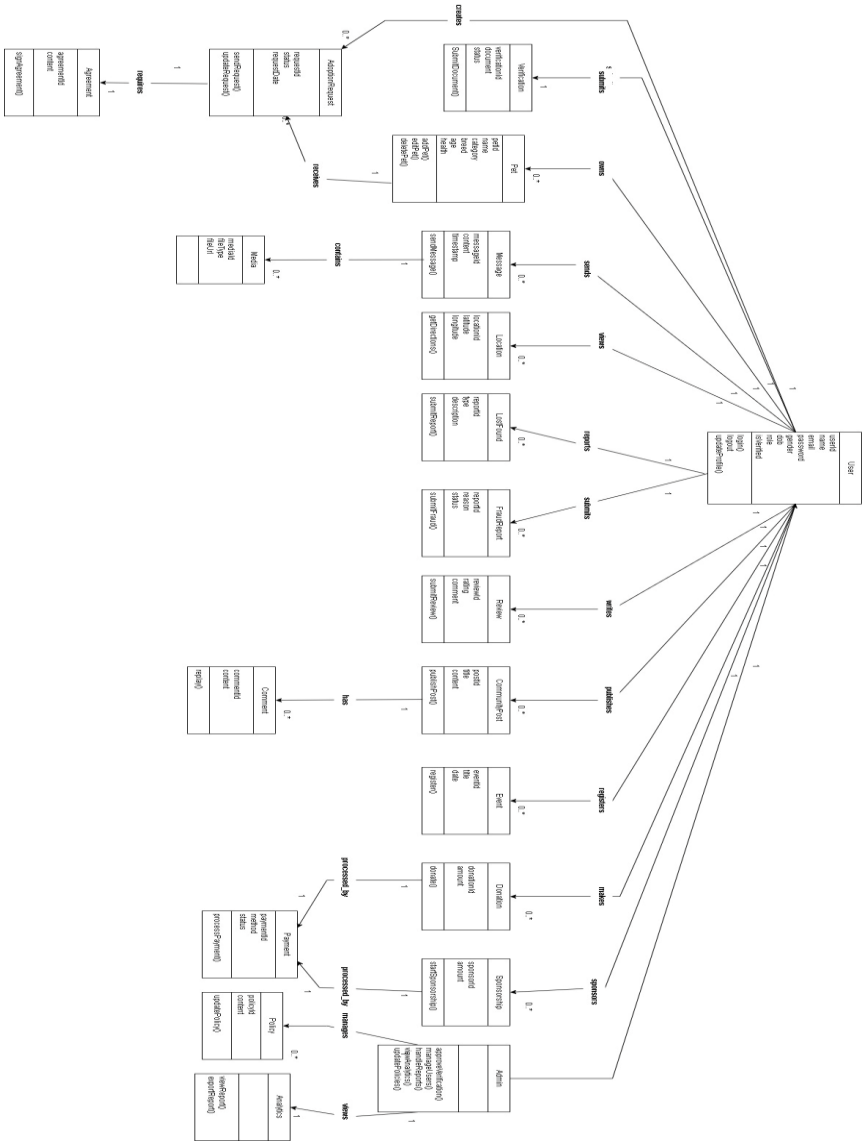
NFR ID	Category	Requirement Description	Priority	Verification Method	Related Feature / Module	Phase
NFR-08	Usability	The system shall provide an intuitive and easy-to-use interface.	High	Usability Testing	All Modules	Phase 1
NFR-09	Usability	The application shall be mobile-responsive for different screen sizes.	High	UI Testing	UI Components	Phase 1
NFR-10	Usability	Error messages and alerts shall be clear and user-friendly.	Medium	Review	Authentication, Forms	Phase 1
NFR-11	Reliability	The system shall maintain data consistency during unexpected failures.	High	Reliability Testing	Database	Phase 2
NFR-12	Reliability	The system shall ensure high availability with minimal downtime.	High	Monitoring	Core System	Phase 2
NFR-13	Reliability	Backup and recovery mechanisms shall prevent data loss.	Medium	Inspection	Database	Phase 2
NFR-14	Scalability	The system shall support future growth in users and pet listings.	High	Review	Overall System	Phase 3

NFR ID	Category	Requirement Description	Priority	Verification Method	Related Feature / Module	Phase
NFR-15	Scalability	New features shall be added without affecting existing functionality.	Medium	Design Review	Modular Architecture	Phase 3
NFR-16	Maintainability	The system shall follow a modular design for easier maintenance.	Medium	Code Review	All Modules	Phase 2
NFR-17	Maintainability	System components shall be properly documented.	Medium	Inspection	Documentation	Phase 2
NFR-18	Compatibility	The system shall be compatible with modern browsers and devices.	High	Compatibility Testing	UI Layer	Phase 1
NFR-19	Compatibility	The system shall integrate smoothly with third-party services.	Medium	Integration Testing	Maps, Payment Gateway	Phase 3
NFR-20	Legal & Compliance	The system shall comply with data privacy and protection policies.	High	Audit	Legal & Policy Module	Phase 3
NFR-21	Legal & Compliance	Adoption agreements and policies shall be stored securely.	Medium	Review	Legal Management	Phase 3

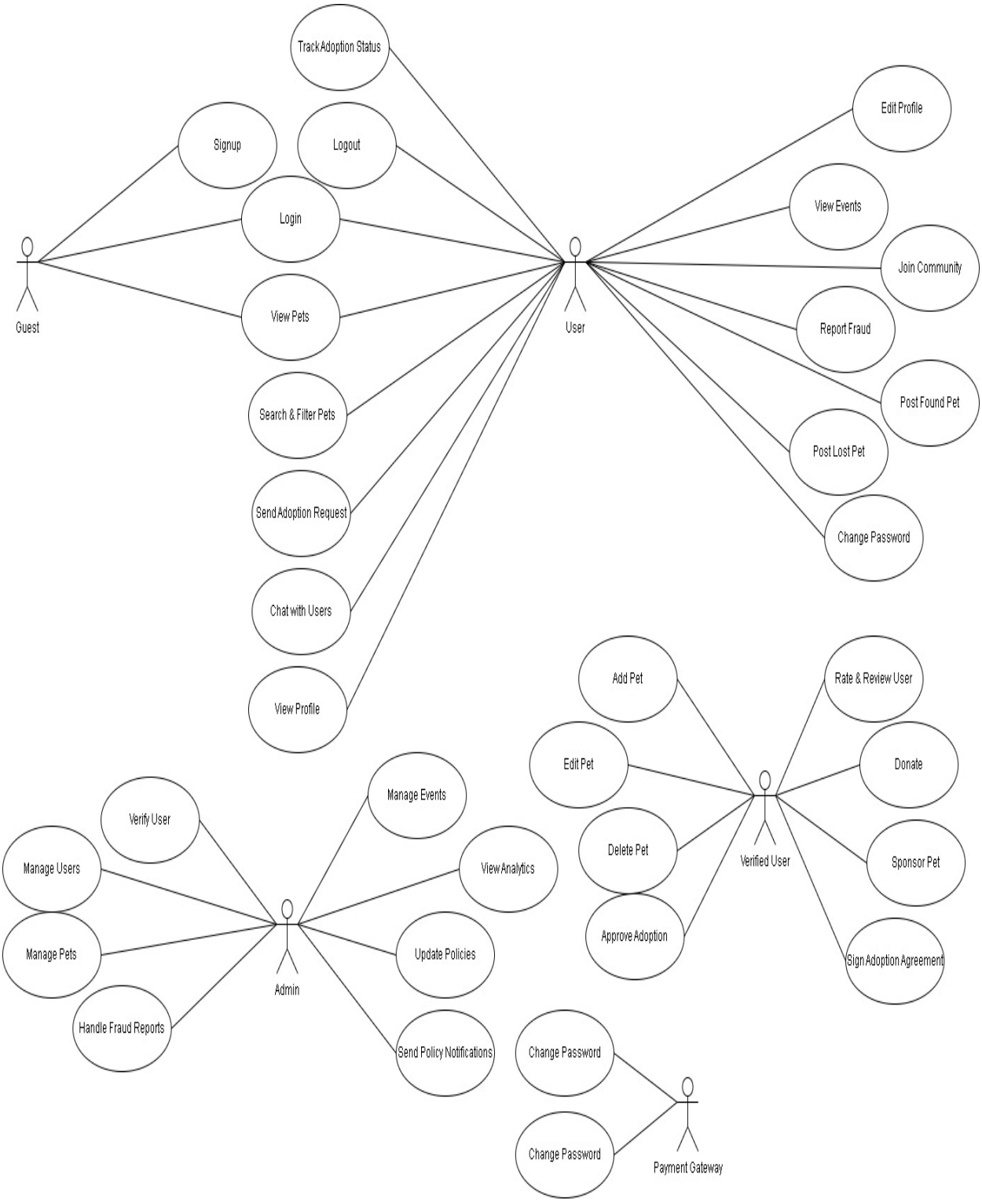
3. SOFTWARE DESIGN

3.1 System Design:

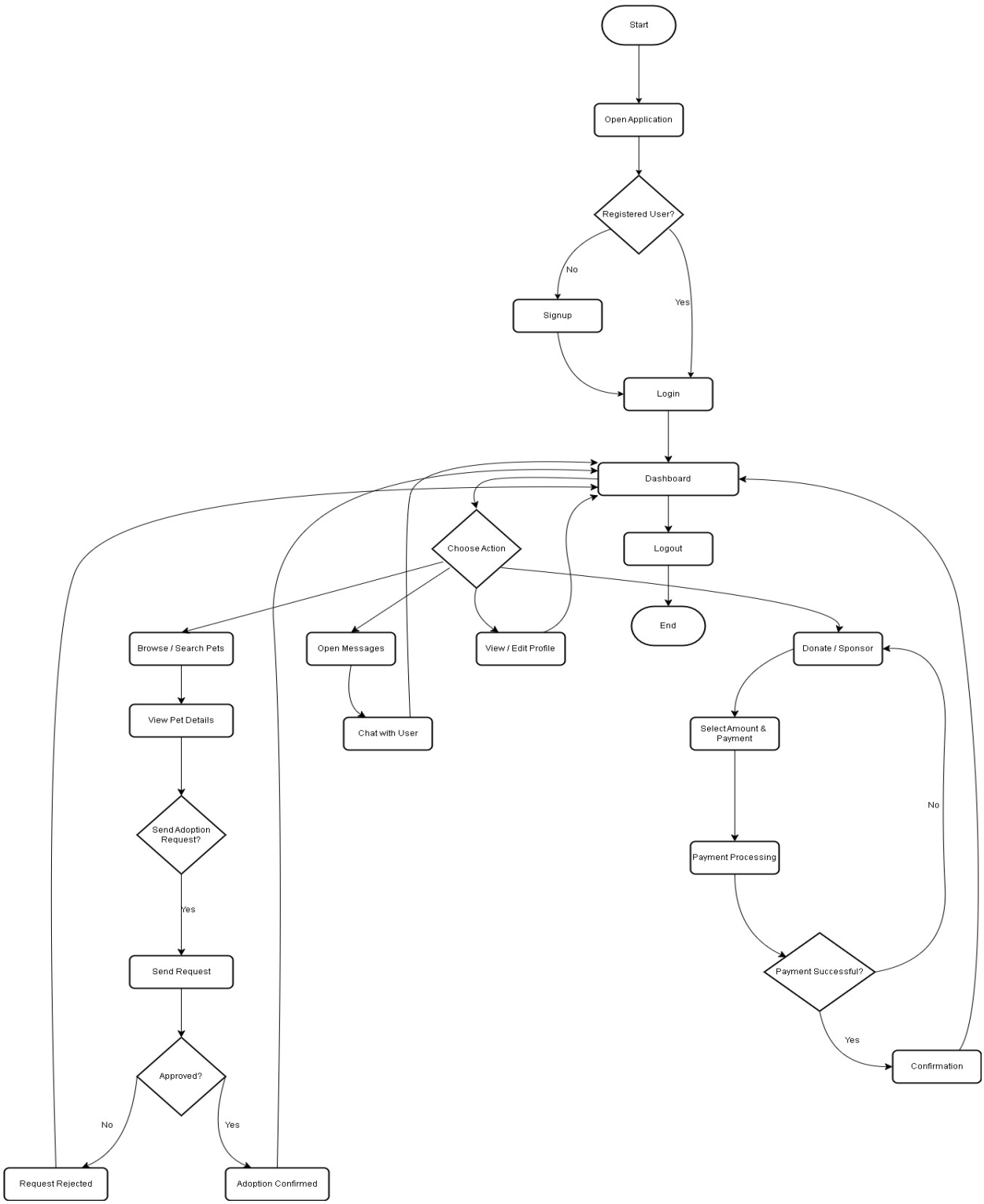
Class Diagram



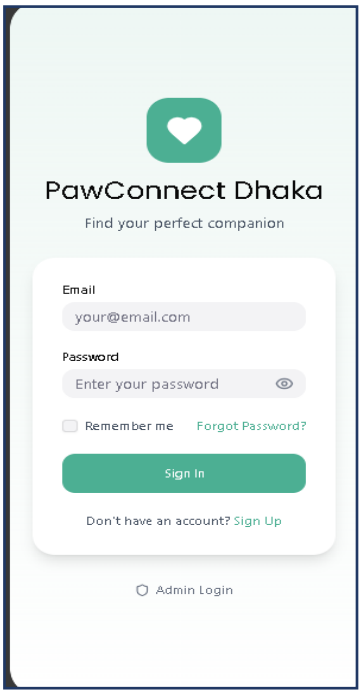
Use Case Diagram

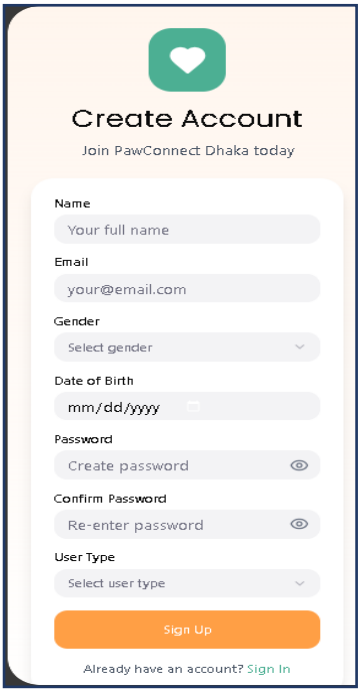


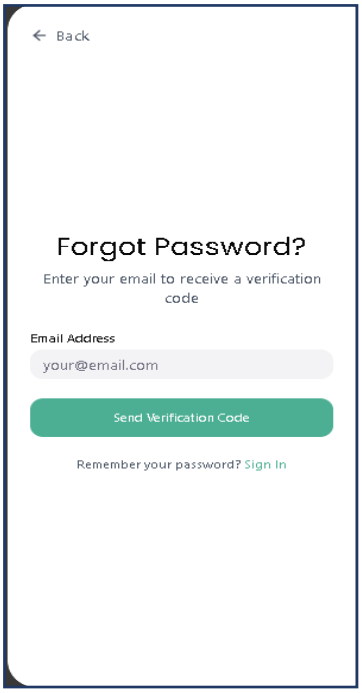
Activity Diagram

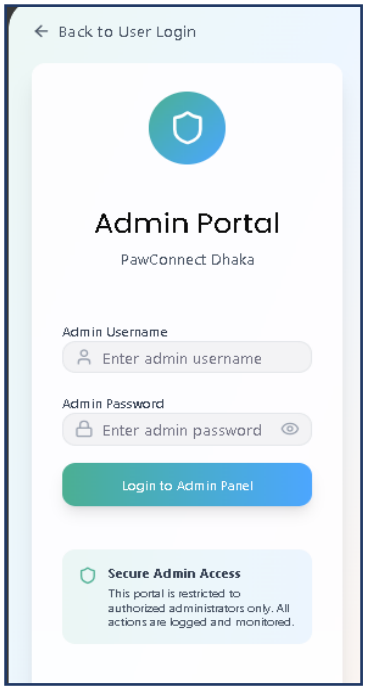


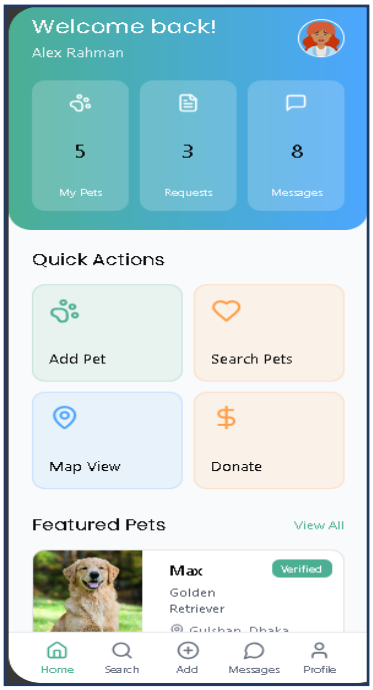
3.2 UI / Wireframe Design using Figma

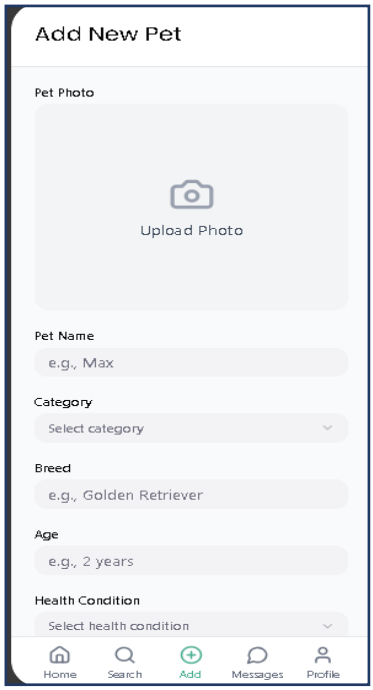


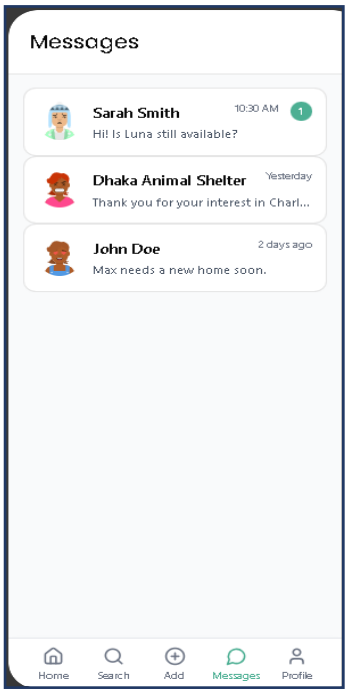


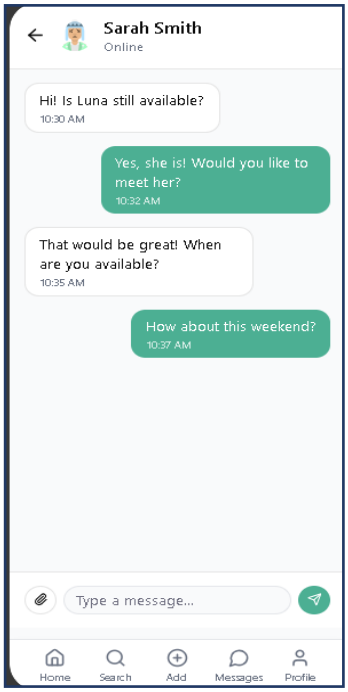


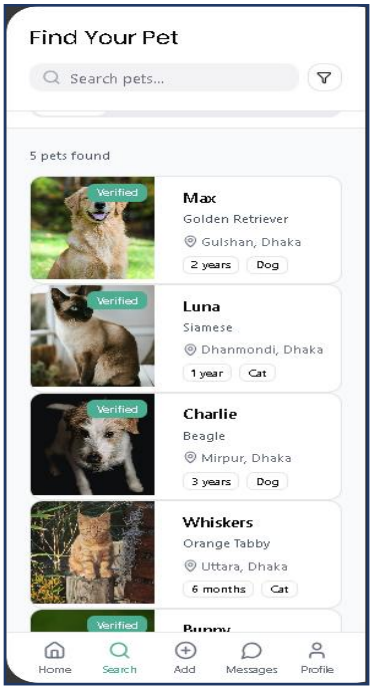


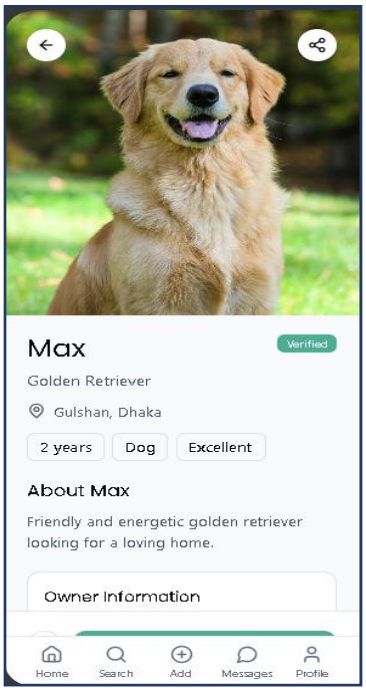

Welcome back!
Alex Rahman
5
My Pets
3
Requests
8
Messages
Quick Actions
Add Pet
Search Pets
Map View
Donate
Featured Pets
View All
Max
Verified
Golden Retriever
Home
Search
Add
Messages
Profile

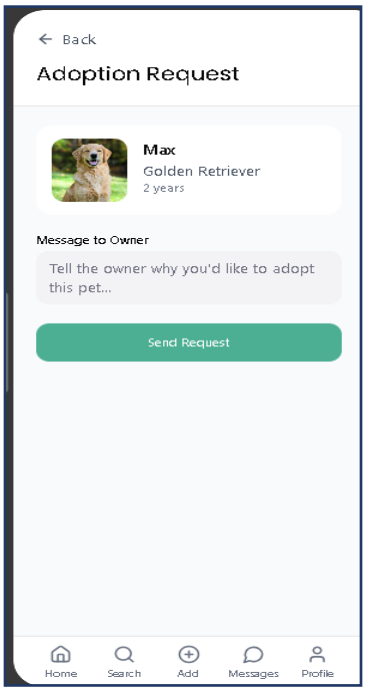

Add New Pet
Pet Photo
Upload Photo
Pet Name
e.g., Max
Category
Select category
Breed
e.g., Golden Retriever
Age
e.g., 2 years
Health Condition
Select health condition
Home
Search
Add
Messages
Profile

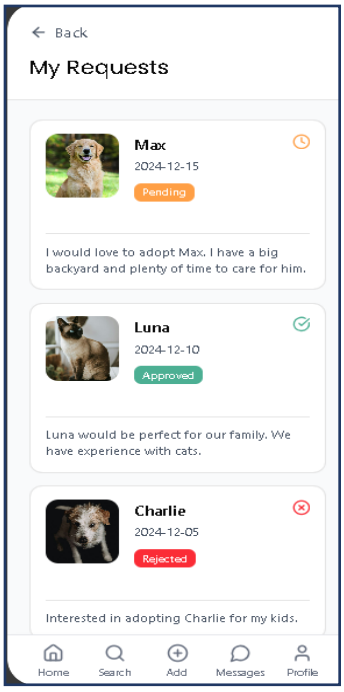

Messages
Sarah Smith
10:30 AM
1
Hi! Is Luna still available?
Dhaka Animal Shelter
Yesterday
Thank you for your interest in Charl...
John Doe
2 days ago
Max needs a new home soon.
Home
Search
Add
Messages
Profile

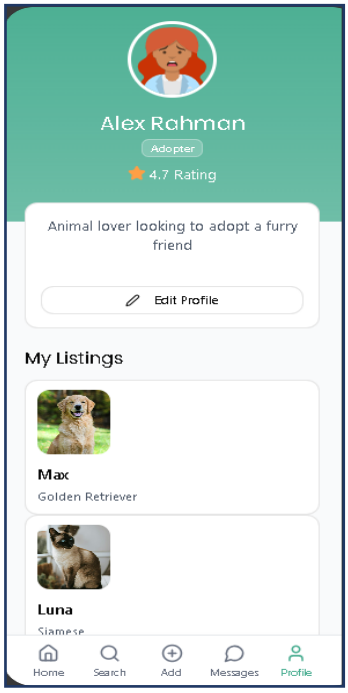

Sarah Smith
Online
Hi! Is Luna still available?
10:30 AM
Yes, she is! Would you like to meet her?
10:32 AM
That would be great! When are you available?
10:35 AM
How about this weekend?
10:37 AM
Type a message...
Home
Search
Add
Messages
Profile


Find Your Pet
Search pets...
5 pets found
Verified
Max
Golden Retriever
Gulshan, Dhaka
2 years
Dog
Verified
Luna
Siamese
Dhanmondi, Dhaka
1 year
Cat
Verified
Charlie
Beagle
Mirpur, Dhaka
3 years
Dog
Whiskers
Orange Tabby
Uttara, Dhaka
6 months
Cat
Verified
Home
Search
Add
Messages
Profile

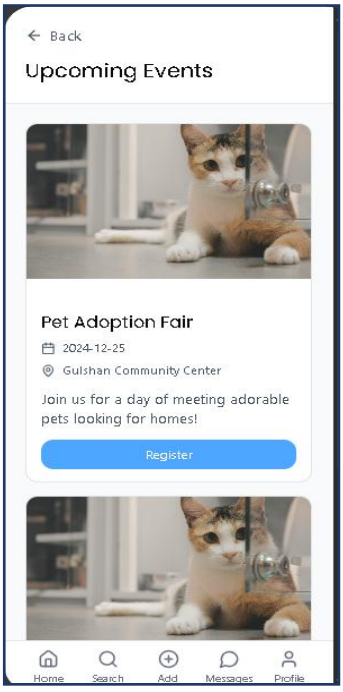

Max
Verified
Golden Retriever
Gulshan, Dhaka
2 years
Dog
Excellent
About Max
Friendly and energetic golden retriever looking for a loving home.
Owner Information
Home
Search
Add
Messages
Profile

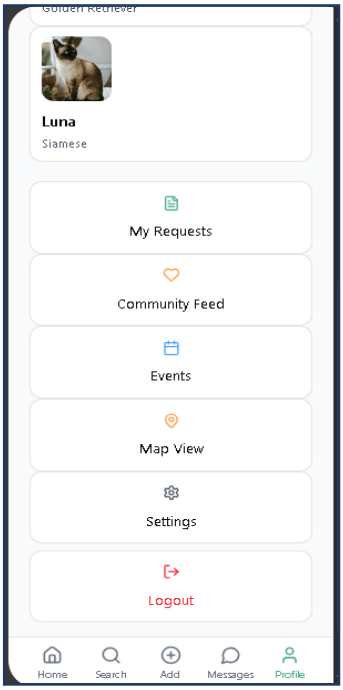

Back
Adoption Request
Max
Golden Retriever
2 years
Message to Owner
Tell the owner why you'd like to adopt this pet...
Send Request
Home
Search
Add
Messages
Profile

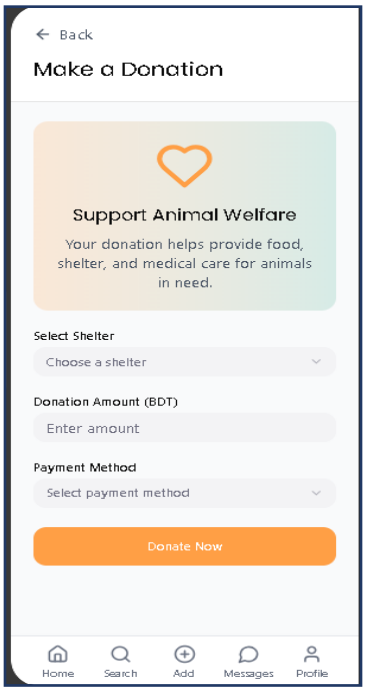

Back
My Requests
Max
2024-12-15
Pending
I would love to adopt Max. I have a big backyard and plenty of time to care for him.
Luna
2024-12-10
Approved
Luna would be perfect for our family. We have experience with cats.
Charlie
2024-12-05
Rejected
Interested in adopting Charlie for my kids.
Home
Search
Add
Messages
Profile

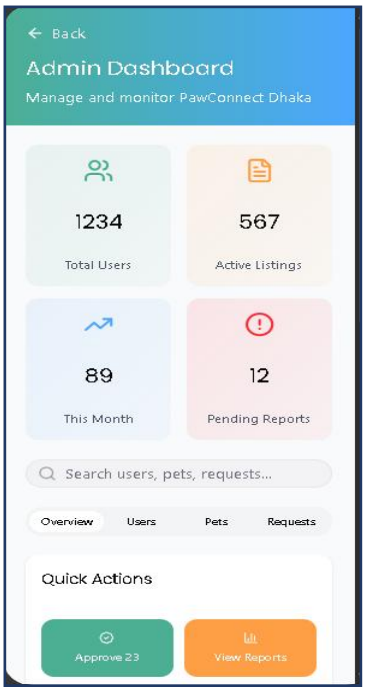

Alex Rahman
Adopter
4.7 Rating
Animal lover looking to adopt a furry friend
Edit Profile
My Listings
Max
Golden Retriever
Luna
Siamese
Home
Search
Add
Messages
Profile

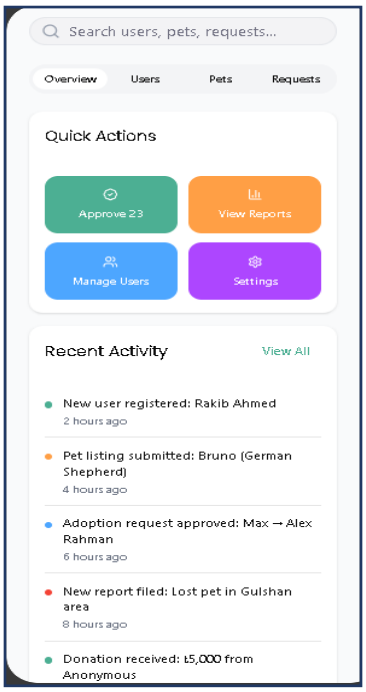

Back
Map View
Map with pins for shelters, clinics, and lost pets
Home
Search
Add
Messages
Profile

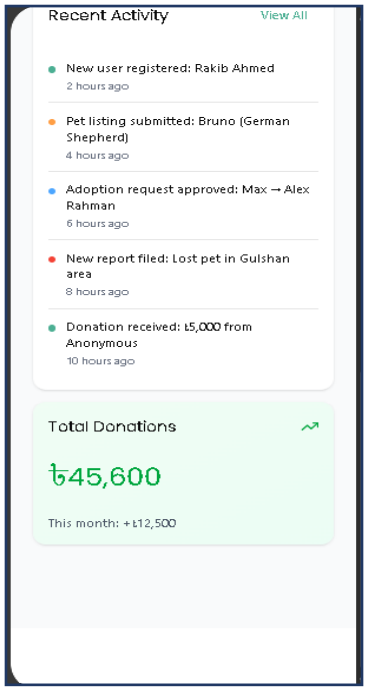

Back
Upcoming Events
Pet Adoption Fair
2024-12-25
Gulshan Community Center
Join us for a day of meeting adorable pets looking for homes!
Register
Home
Search
Add
Messages
Profile

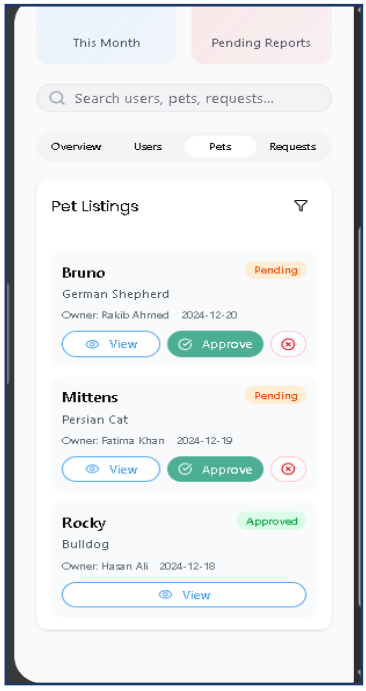

Luna
Siamese
My Requests
Community Feed
Events
Map View
Settings
Logout
Home
Search
Add
Messages
Profile

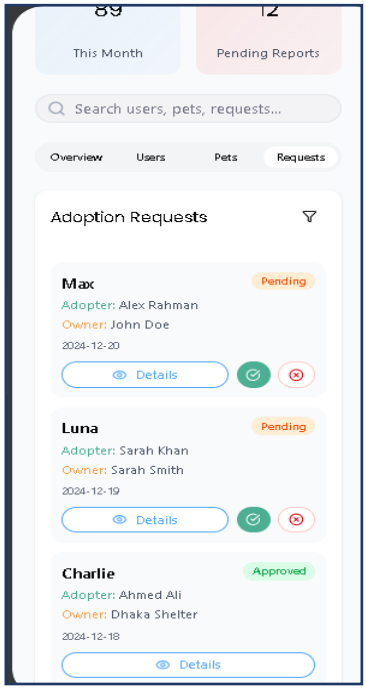

Back
Make a Donation
Support Animal Welfare
Your donation helps provide food, shelter, and medical care for animals in need.
Select Shelter
Choose a shelter
Donation Amount (BDT)
Enter amount
Payment Method
Select payment method
Donate Now
Home
Search
Add
Messages
Profile

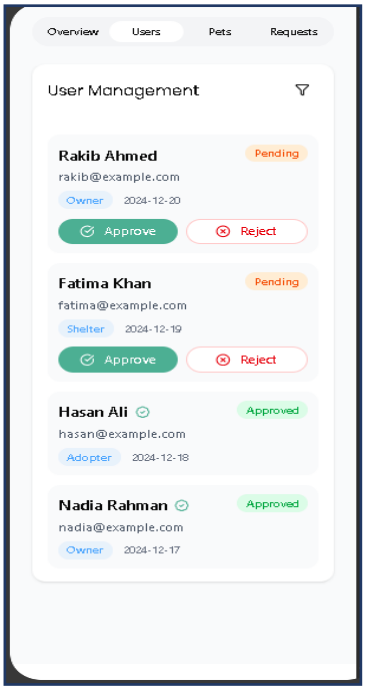

Back
Admin Dashboard
Manage and monitor PawConnect Dhaka
1234
Total Users
567
Active Listings
89
This Month
12
Pending Reports
Search users, pets, requests...
Overview
Users
Pets
Requests
Quick Actions
Approve 23
View Reports


Search users, pets, requests...
Overview
Users
Pets
Requests
Quick Actions
Approve 23
View Reports
Manage Users
Settings
Recent Activity
View All
New user registered: Rakib Ahmed
2 hours ago
Pet listing submitted: Bruno (German Shepherd)
4 hours ago
Adoption request approved: Max → Alex Rahman
6 hours ago
New report filed: Lost pet in Gulshan area
8 hours ago
Donation received: ৳5,000 from Anonymous

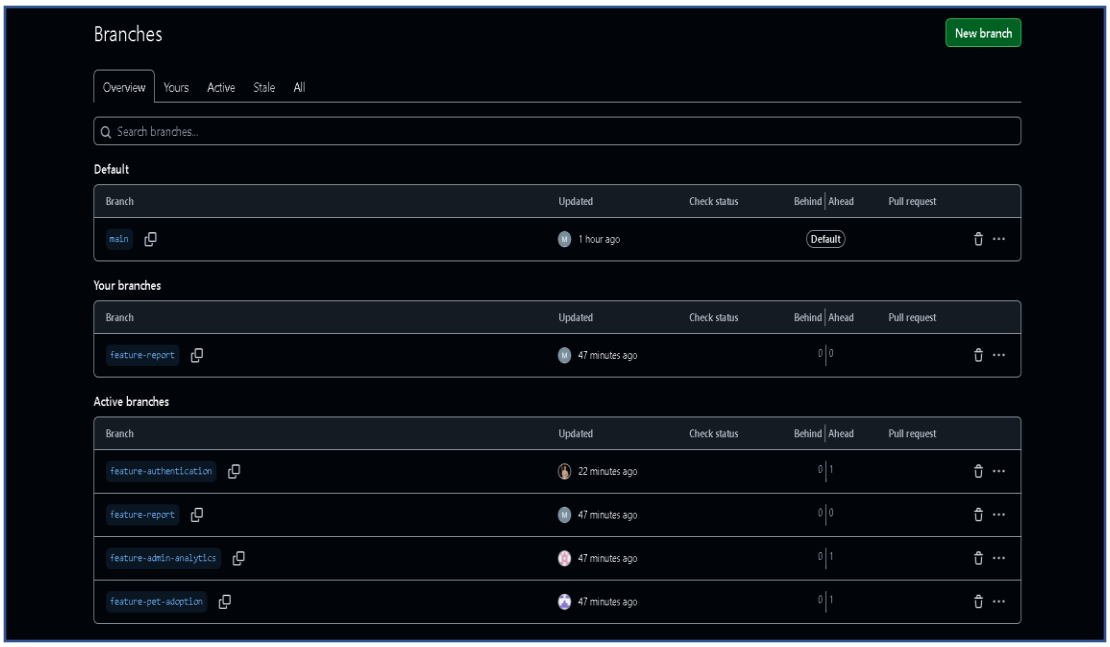

Recent Activity
View All
New user registered: Rakib Ahmed
2 hours ago
Pet listing submitted: Bruno (German Shepherd)
4 hours ago
Adoption request approved: Max → Alex Rahman
6 hours ago
New report filed: Lost pet in Gulshan area
8 hours ago
Donation received: ৳5,000 from Anonymous
10 hours ago
Total Donations
৳45,600
This month: +৳12,500

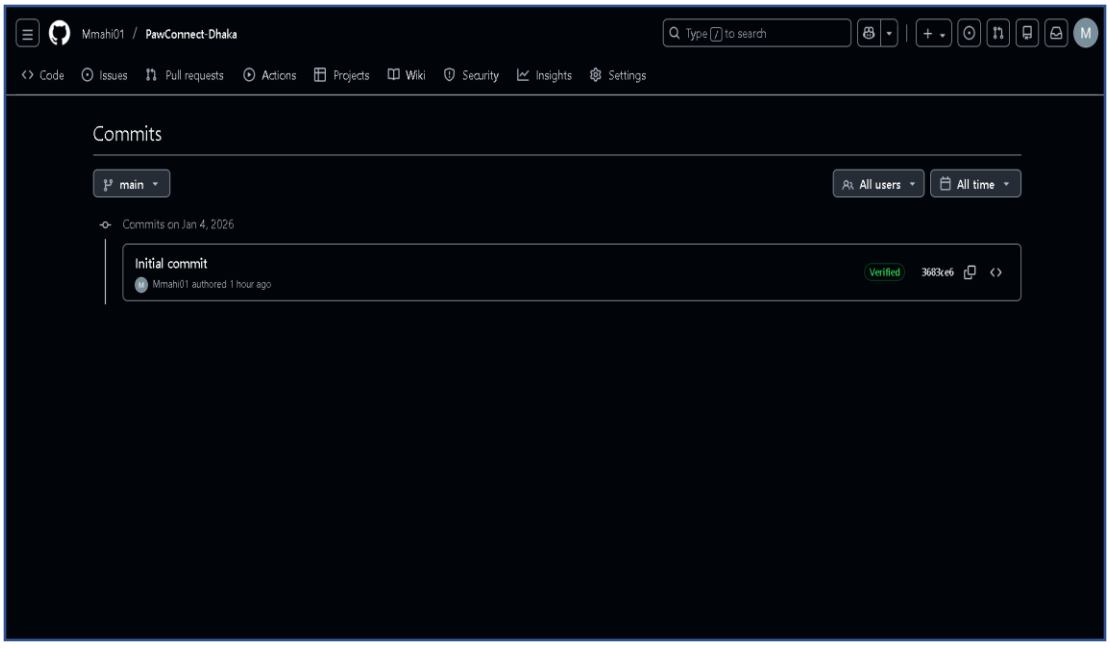

This Month
Pending Reports
Search users, pets, requests...
Overview
Users
Pets
Requests
Pet Listings
Bruno
Pending
German Shepherd
Owner: Rakib Ahmed 2024-12-20
View
Approve
Mittens
Pending
Persian Cat
Owner: Fatima Khan 2024-12-19
View
Approve
Rocky
Approved
Bulldog
Owner: Hasan Ali 2024-12-18
View


89
This Month
12
Pending Reports
Search users, pets, requests...
Overview
Users
Pets
Requests
Adoption Requests
Max
Pending
Adopter: Alex Rahman
Owner: John Doe
2024-12-20
Details
Luna
Pending
Adopter: Sarah Khan
Owner: Sarah Smith
2024-12-19
Details
Charlie
Approved
Adopter: Ahmed Ali
Owner: Dhaka Shelter
2024-12-18
Details


Overview
Users
Pets
Requests
User Management
Rakib Ahmed
Pending
rakib@example.com
Owner
2024-12-20
Approve
Reject
Fatima Khan
Pending
fatima@example.com
Shelter
2024-12-19
Approve
Reject
Hasan Ali
Approved
hasan@example.com
Adopter
2024-12-18
Nadia Rahman
Approved
nadia@example.com
Owner
2024-12-17

4. GIT WORKFLOW





5. SOFTWARE TESTING

Project Name : PawConnect Dhaka	**Test Designed by: Mahfuzur Rahman Khan Fahad**
Test Case: TC_01	Test Designed Date:30/12/2025
Test Priority: High	Test Executed by: Mahfuzur Rahman Khan Fahad
Module Name: Authentication	Test Execution Date:30/12/2025
Test Title: Verify login with valid email and password	
Description: Test the login functionality	
Precondition: User has a registered account	
Dependencies: Internet connection	

Test Steps	Test Data	Expected Results	Actual Results	Status
1. Go to login Page 2. Enter email 3. Enter Password 4. Click Signin	Email:user@mail.com Password: ****	User is logged in successfully	**As expected**	**Pass**

Project Name : PawConnect Dhaka	**Test Designed by: Mahfuzur Rahman Khan Fahad**
Test Case: TC_02	Test Designed Date:30/12/2025
Test Priority: Medium	Test Executed by: Mahfuzur Rahman Khan Fahad
Module Name: Authentication	Test Execution Date:30/12/2025
Test Title: Verify logout functionality	
Description: Test logout process	
Precondition: User is logged in	
Dependencies: Active session	

Test Steps	Test Data	Expected Results	Actual Results	Status
1. Click logout button 2. Confirm logout	**Yes**	User redirected to login page	**As expected**	**Pass**

Project Name : PawConnect Dhaka	**Test Designed by: Mahfuzur Rahman Khan Fahad**
Test Case: TC_03	Test Designed Date:30/12/2025
Test Priority: Medium	Test Executed by: Mahfuzur Rahman Khan Fahad
Module Name: Authentication	Test Execution Date:30/12/2025
Test Title: Verify login with invalid password	
Description: Test login failure scenario	
Precondition: User is registered	
Dependencies: User account exists	

Test Steps	Test Data	Expected Results	Actual Results	Status
1. Enter valid email 2. Enter wring Password	Email:user@mail.com Password: wrong123	Error message displayed	**As expected**	**Pass**

Project Name : PawConnect Dhaka	**Test Designed by: Moynul Islam Mahi**
Test Case: TC_04	Test Designed Date:30/12/2025
Test Priority: High	Test Executed by: Moynul Islam Mahi
Module Name: Pet Management	Test Execution Date:30/12/2025
Test Title: Verify add pet listing	
Description: Test pet listing creation	
Precondition: User logged in	
Dependencies: Verified user	

Test Steps	Test Data	Expected Results	Actual Results	Status
1. Open add pet page 2. Enter pet details 3. Submit report	Name, Breed, Age	Pet listed successfully	**As expected**	**Pass**

Project Name : PawConnect Dhaka	**Test Designed by: Moynul Islam Mahi**
Test Case: TC_05	Test Designed Date:30/12/2025
Test Priority: Medium	Test Executed by: Moynul Islam Mahi
Module Name: Pet Management	Test Execution Date:30/12/2025
Test Title: Verify edit pet details	
Description: Test editing pet info	
Precondition: Pet listing exist	
Dependencies: Pet ID available	

Test Steps	Test Data	Expected Results	Actual Results	Status
1. Select Pet 2. Update Details	Pet ID:P101 Age Updated	Changes Saved	**As expected**	**Pass**

Project Name : PawConnect Dhaka	**Test Designed by: Moynul Islam Mahi**
Test Case: TC_06	Test Designed Date:30/12/2025
Test Priority: Medium	Test Executed by: Moynul Islam Mahi
Module Name: Pet Management	Test Execution Date:30/12/2025
Test Title: Verify delete pet listing	
Description: Test pet deletion	
Precondition: Pet listing exists	
Dependencies: Pet ID available	

Test Steps	Test Data	Expected Results	Actual Results	Status
1. Click delete 2. Confirm delete	**Pet ID: P101** Yes	Pet removed	**Pet still available**	**Fail**

Project Name : PawConnect Dhaka	**Test Designed by: Sanzida Aruba Shorony**
Test Case: TC_07	Test Designed Date:30/12/2025
Test Priority: Medium	Test Executed by: Sanzida Aruba Shorony
Module Name: Adoption	Test Execution Date:30/12/2025
Test Title: Verify adoption request submission	
Description: Test adoption request process	
Precondition: User logged in & pet available	
Dependencies: Pet listing exists	

Test Steps	Test Data	Expected Results	Actual Results	Status
1. Open pet details 2. Click Adopt 3. Submit request	Pet ID: P123 Message Text	Request sent	**As expected**	**Pass**

Project Name : PawConnect Dhaka	**Test Designed by: Sanzida Aruba Shorony**
Test Case: TC_08	Test Designed Date:30/12/2025
Test Priority: Medium	Test Executed by: Sanzida Aruba Shorony
Module Name: Adoption	Test Execution Date:30/12/2025
Test Title: Verify adoption request status update	
Description: Test request tracking	
Precondition: Request submitted	
Dependencies: Adoption request exists	

Test Steps	Test Data	Expected Results	Actual Results	Status
1. Open Requests Page 2. Check status	Pending	Status shown correctly	**As expected**	**Pass**

Project Name : PawConnect Dhaka	**Test Designed by: Sanzida Aruba Shorony**
Test Case: TC_09	Test Designed Date:30/12/2025
Test Priority: Medium	Test Executed by: Sanzida Aruba Shorony
Module Name: Adoption	Test Execution Date:30/12/2025
Test Title: Verify adoption request cancellation	
Description: Test cancel request feature	
Precondition: Pending request exists	
Dependencies: Adoption request exists	

Test Steps	Test Data	Expected Results	Actual Results	Status
1. Click cancel request	Request ID	Request cancelled	**Error shown**	**Fail**

Project Name : PawConnect Dhaka	**Test Designed by: MD. Jahidul Islam**
Test Case: TC_10	Test Designed Date:30/12/2025
Test Priority: Medium	Test Executed by: MD. Jahidul Islam
Module Name: Messaging	Test Execution Date:30/12/2025
Test Title: Verify send message	
Description: Test chat functionality	
Precondition: Both users verified	
Dependencies: Active chat session	

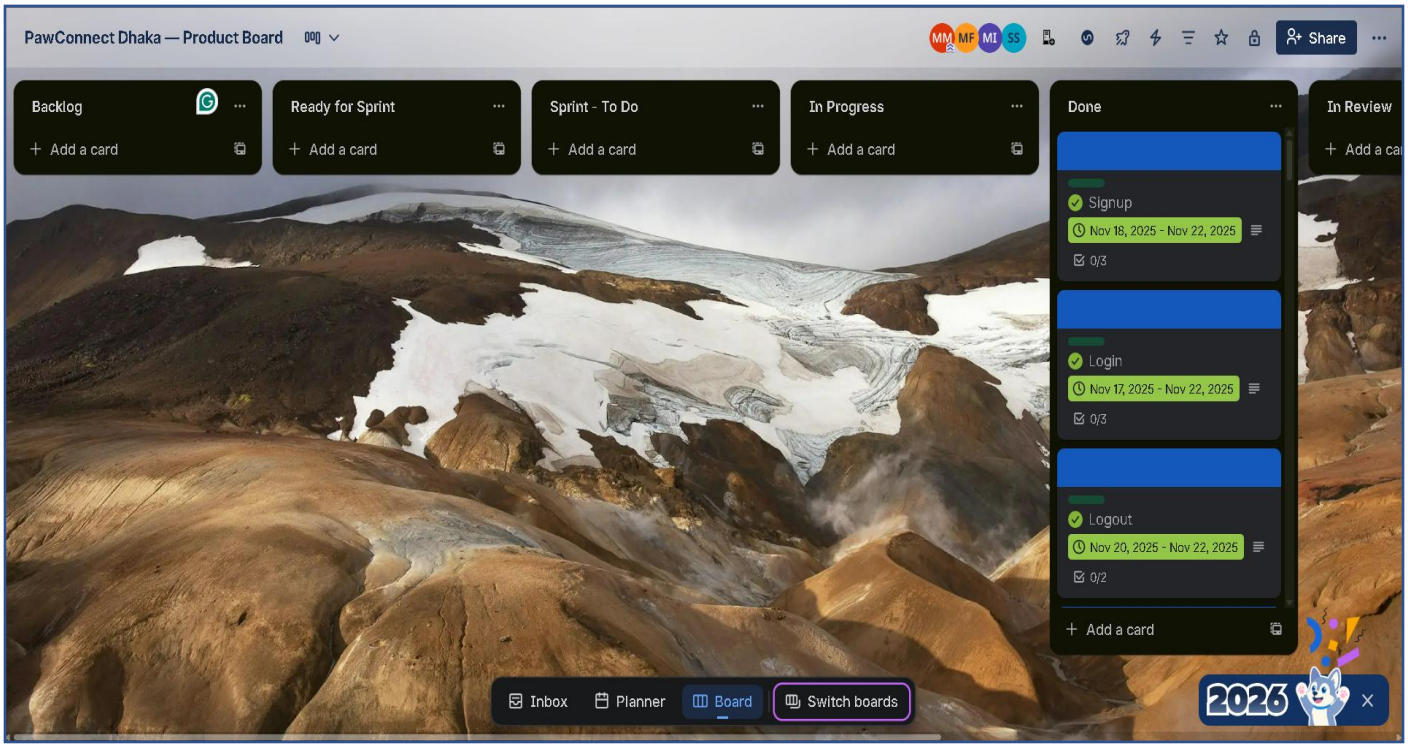
Test Steps	Test Data	Expected Results	Actual Results	Status
1. Open Chat 2. Send message	"Hello"	Message Delivered	**As expected**	**Pass**

Project Name : PawConnect Dhaka	**Test Designed by: MD. Jahidul Islam**
Test Case: TC_11	Test Designed Date:30/12/2025
Test Priority: Medium	Test Executed by: MD. Jahidul Islam
Module Name: Profile	Test Execution Date:30/12/2025
Test Title: Verify update profile	
Description: Test profile update	
Precondition: User logged in	
Dependencies: Profile exists	

Test Steps	Test Data	Expected Results	Actual Results	Status
1. Open profile 2. Update bio	New bio	Changes saved	**As expected**	**Pass**

Project Name : PawConnect Dhaka	**Test Designed by: MD. Jahidul Islam**
Test Case: TC_12	Test Designed Date:30/12/2025
Test Priority: Medium	Test Executed by: MD. Jahidul Islam
Module Name: Profile	Test Execution Date:30/12/2025
Test Title: Verify profile photo upload	
Description: Test photo upload	
Precondition: User logged in	
Dependencies: Storage available	

Test Steps	Test Data	Expected Results	Actual Results	Status
1. Upload image	Large image file	Image Uploaded	**Upload failed**	**Fail**


PawConnect Dhaka — Product Board
Share
Backlog
Add a card
Ready for Sprint
Add a card
Sprint - To Do
Add a card
In Progress
Add a card
Done
In Review
Signup
Nov 18, 2025 - Nov 22, 2025
0/3
Login
Nov 17, 2025 - Nov 22, 2025
0/3
Logout
Nov 20, 2025 - Nov 22, 2025
0/2
Add a card
Inbox
Planner
Board
Switch boards
2026

6. CONCLUSION

The PawConnect Dhaka project was designed to address the growing need for a secure, reliable, and community-driven digital platform for pet adoption, care, and welfare management in Dhaka. By integrating features such as user verification, pet listing and adoption management, real-time messaging, location-based services, and community engagement, the system ensures transparency, trust, and ease of use for all stakeholders. The inclusion of administrative controls, fraud reporting, analytics, and legal policy management further strengthens the platform's reliability and compliance.

Throughout the development planning, the Agile (Scrum) process model was selected to support iterative development, continuous feedback, and adaptability to changing requirements. This approach allowed the project to be broken into manageable phases, ensuring timely delivery of core functionalities while maintaining flexibility for future enhancements. Proper documentation through the PRD, test cases, and traceability matrices ensured alignment between requirements, design, and implementation.

In addition, collaborative tools such as GitHub facilitated effective teamwork, version control, and transparency among team members. Overall, PawConnect Dhaka demonstrates how thoughtful system design, structured development practices, and user-centric features can contribute to a meaningful technological solution that supports animal welfare, responsible adoption, and community awareness in an urban environment.